Partial-Label Metric Ceilings for GRN Evaluation

A Theory-First, Adversarially Stress-Tested Study

Abstract

Gene regulatory network (GRN) benchmarks are usually interpreted as if labels were complete, but curated references are partial observations of latent biology. We formalize this partial-label regime and derive explicit ceilings for observed F1 and AUPR as functions of positive-label coverage `c`. We then execute a full low-compute analysis on existing benchmark artifacts, propagate uncertainty in `c`, and adversarially stress-test assumptions using non-MAR missingness simulations. Across `39` AUPR-evaluable rows (15 methods, 5 references), the best ceiling-normalized scores are modest (best F1 ratio `0.137`, best AUPR ratio `0.0138`), with medians near zero (F1 ratio `0.000000`, AUPR ratio `0.000056`). MAR-theory simulations show low absolute bias (median F1 bias `7.298e-04`, median AUPR bias `1.346e-03`), while non-MAR missingness can induce substantial AUPR deviations (optimistic labeling: +`0.090`; pessimistic: `-0.073` vs MAR theory). We provide a reproducible ceiling-aware reporting framework and a finalized artifact bundle.

1. Introduction

Absolute benchmark metrics in GRN inference are hard to interpret under incomplete references. If only a fraction of true edges are labeled positive, observed precision-style metrics are systematically distorted, and even an ideal predictor can appear weak in absolute terms. This paper answers Proposal 1's central question directly: **what observed F1/AUPR is theoretically achievable under partial positive labels, and how far are current results from that ceiling?**

We contribute:

1. Closed-form ceiling formulas for observed precision/F1/AUPR under a missing-at-random (MAR) positive-label model.
2. Coverage-aware reinterpretation of existing outputs, including uncertainty bands and adversarial sensitivity scenarios.
3. Synthetic theorem validation and adversarial non-MAR tests that quantify where ceiling assumptions hold or fail.
4. A full reproducible paper package with tables, figures, and manuscript-ready outputs.

2. Formal Setup

Let:

- `U`: candidate edge universe.
- `T subset U`: latent true positives.
- `L+ subset T`: observed positives in a reference.
- `c = P(e in L+ | e in T)`: positive-label coverage.
- `P`: predicted positives for a method.

Define latent and observed metrics:

- `Prec_lat = |P intersect T| / |P|`, `Rec_lat = |P intersect T| / |T|`.
- `Prec_obs = |P intersect L+| / |P|`, `Rec_obs = |P intersect L+| / |L+|`.

Assumption A1 (MAR): each latent true edge is observed independently with probability `c`.

3. Theory

Proposition 1: Observed-latent relation

Under A1:

- `E[Prec_obs] = c * Prec_lat`

- `E[Rec_obs] = Rec_lat`

So latent precision can be estimated by `Prec_obs / c` when `c` is estimated.

Proposition 2: Observed F1 ceiling

For an ideal latent predictor (`Prec_lat = Rec_lat = 1`):

- `F1_obs,max(c) = 2c / (1 + c)`.

Proposition 3: Observed AUPR ceiling

For latent positive count `t = |T|` and ideal latent ranking:

- `AUPR_obs,max(c, t) ~= c + (1-c) * H_t/t`, where `H_t` is harmonic.
- Large-`t` limit: `AUPR_obs,max ~= c`.

Observed random baseline under partial labels:

- `AUPR_rand,obs = c * b` where `b = |T|/|U|`.

Normalized headroom statistic:

- `rho_aupr = (AUPR_obs - AUPR_rand,obs) / (AUPR_obs,max - AUPR_rand,obs)`.

Coverage uncertainty model

We model reference-level coverage as:

- `c ~ Beta(k+1, n-k+1)` with overlap support `k` and denominator `n`, then propagate uncertainty by Monte Carlo draws.

4. Data and Protocol

4.1 Primary artifacts

- `network_inference/outputs/score_eval_probe_priors.csv`
- `network_inference/outputs/score_eval_grn_baselines_immune.csv`

4.2 F1-extended artifacts

- `network_inference/outputs/score_eval_probe_priors_full_genes.csv`
- `network_inference/outputs/score_eval_probe_priors_full_genes_crosswalk.csv`
- `network_inference/outputs/score_eval_probe_priors_full_genes_omnipath.csv`

4.3 Coverage proxy artifacts

- `network_inference/outputs/score_eval_probe_priors_missing_report.json`
- `network_inference/outputs/score_eval_probe_priors_full_genes_missing_report.json`
- `network_inference/outputs/score_eval_probe_priors_full_genes_crosswalk_missing_report.json`
- `network_inference/outputs/score_eval_probe_priors_full_genes_omnipath_missing_report.json`

4.4 Coverage scenarios

For each reference, we evaluate:

- `overlap_point` (central).
- `beta_ci_low`, `beta_ci_high`.
- `adversarial_half`, `adversarial_150pct`.
- `mapping_min`, `mapping_max` (when available).
- `conservative_min`, `optimistic_max` envelope.

4.5 Synthetic experiments

- MAR simulations over grids of `c in {0.05,0.1,0.2,0.4,0.7,0.9}`, `t in {50,200,1000,5000}`.
- Non-MAR stress test with optimistic/pessimistic rank-dependent labeling.

5. Results

5.1 Core reinterpretation

- AUPR-evaluable rows: `39`
- F1-evaluable rows: `117`
- Methods: `15`
- References: `5`

Key extrema:

- Best F1 ceiling ratio: `genie3` on `dorothea_trrust_union_immune` = `0.137393`
- Best AUPR ceiling ratio: `grnboost2` on `dorothea_trrust_union_immune` = `0.013771`
- Median F1 ceiling ratio: `0.000000`
- Median AUPR ceiling ratio: `0.000056`
- Median normalized AUPR uplift ratio: `-2.904159e-06`

Table 1. Reference-level ceiling summary

reference	n_rows	methods	c_median	observed_f1_median	f1_ceiling_median	f1_ratio
dorothea_trrust_union_immune	9	9	0.117816	0.00217946	0.210797	0
dorothea_human	6	6	0.094976	0.000178214	0.173476	0.
trrust_human	6	6	0.159027	0	0.274414	
hpn_dream	9	9	0.745098	0	0.853933	
beeline_gsd	9	9	0.882353	0	0.9375	

Interpretation: references with low overlap-based coverage (`dorothea_human`, `dorothea_trrust_union_immune`) have lower absolute ceilings, but observed metrics still remain small fractions of those ceilings.

5.2 Method-level normalized rankings

Table 2. Method summary (raw vs normalized)

method	observed_aupr_median	aupr_ratio_median	rank_observed_aupr	rank_aupr_ratio
probe_consensus	0.000159345	0.00165566	1	1
probe_perturbation	0.000144556	0.00150017	3	2
probe_integrated_gradients	0.000145163	0.00148826	2	3
probe_grad_input	0.00014285	0.00148224	4	4
probe_attention	0.000122129	0.00126438	5	5
attention_inferred	0.000121964	0.00126265	6	6
grnboost2	3.01513e-06	4.00071e-06	7	7
genie3	3.01509e-06	4.00065e-06	8	8
scenic_grnboost2	3.01484e-06	4.00032e-06	9	9
scenic_pruned	3.01475e-06	4.0002e-06	10	10
pidc_full	3.01472e-06	4.00017e-06	11	11
pidc_proxy	3.01472e-06	4.00016e-06	12	12
spearman	3.01439e-06	3.99972e-06	13	13
pearson	3.01439e-06	3.99972e-06	14	14
random	3.0138e-06	3.99893e-06	15	15

Rank-shift finding: `probe_perturbation` moved up by 1 rank position(s) (raw rank 3 -> normalized rank 2). `probe_integrated_gradients` moved down by 1 rank position(s) (raw rank 2 -> normalized rank 3).

5.3 Coverage-scenario stability

Table 3. Scenario-level rank stability

scenario	n_methods	spearman_vs_overlap_rank	spearman_vs_raw_rank	max_abs_rank_shift_vs_ov
adversarial_150pct	15	1	0.996429	
adversarial_half	15	1	0.996429	
beta_ci_high	15	1	0.996429	
beta_ci_low	15	1	0.996429	
conservative_min	15	1	0.996429	
mapping_max	9	0.5	0.5	
mapping_min	9	0.55	0.55	
optimistic_max	15	1	0.996429	
overlap_point	15	1	0.996429	

Interpretation:

- Overlap/Beta/adversarial scaling scenarios preserve ordering (Spearman ~1.0).
- Mapping-driven scenarios (`mapping_min/max`) induce major shifts but only on a subset of methods/references where those proxies exist.

5.4 Adversarial coverage sensitivity

Table 4. Most coverage-sensitive methods (top 8 by AUPR-ratio range)

method	aupr_ratio_min	aupr_ratio_median	aupr_ratio_max	aupr_ratio_range	f1_ratio_min	f1_
grnboost2	1.23789e-06	7.0927e-06	0.0275386	0.0275374	0	
genie3	1.2379e-06	7.09267e-06	0.0221169	0.0221157	0	
scenic_grnboost2	1.23865e-06	7.09456e-06	0.00770431	0.00770307	0	
probe_consensus	3.71711e-05	0.00114075	0.00651042	0.00647324	0	
pidc_proxy	1.2388e-06	7.09485e-06	0.00613842	0.00613718	0	
probe_perturbation	3.71714e-05	0.00103707	0.00588855	0.00585137	0	
spearman	1.23786e-06	7.09175e-06	0.0058233	0.00582206	0	
pearson	1.23786e-06	7.09175e-06	0.00580037	0.00579913	0	

Coverage sensitivity is highly method-dependent; some methods show large ceiling-ratio range under plausible-to-adversarial `c` bands.

5.5 MAR theorem validation

Table 5. Synthetic MAR validation (bias relative to theory)

c	t	f1_obs_mean	f1_theory	f1_bias_abs	aupr_obs_mean	aupr_theory	aupr_bias_abs
0.05	50	0.100644	0.0952381	0.00540598	0.127244	0.135485	0.0082409
0.05	200	0.0932589	0.0952381	0.00197922	0.0719671	0.0779206	0.0059535
0.05	1000	0.095733	0.0952381	0.000494944	0.0573385	0.0571112	0.000227278
0.05	5000	0.0949053	0.0952381	0.000332764	0.0512554	0.051728	0.000472586
0.1	50	0.180304	0.181818	0.0015142	0.155571	0.180986	0.0254151

c	t	f1_obs_mean	f1_theory	f1_bias_abs	aupr_obs_mean	aupr_theory	aupr_bias_abs
0.1	200	0.182858	0.181818	0.00103984	0.123486	0.126451	0.00296543
0.1	1000	0.180888	0.181818	0.000929746	0.10488	0.106737	0.00185676
0.1	5000	0.181712	0.181818	0.000106567	0.101468	0.101637	0.000168657
0.2	50	0.32473	0.333333	0.00860329	0.254503	0.271987	0.0174844
0.2	200	0.332166	0.333333	0.00116688	0.219643	0.223512	0.00386902
0.2	1000	0.33163	0.333333	0.00170304	0.204336	0.205988	0.00165211
0.2	5000	0.333516	0.333333	0.000183104	0.201402	0.201455	5.2627e-05
0.4	50	0.563338	0.571429	0.00809046	0.434901	0.45399	0.0190897
0.4	200	0.569741	0.571429	0.00168727	0.412161	0.417634	0.005473
0.4	1000	0.571482	0.571429	5.37676e-05	0.405464	0.404491	0.000972989
0.4	5000	0.571631	0.571429	0.000201988	0.40085	0.401091	0.000241273
0.7	50	0.82088	0.823529	0.00264951	0.723628	0.726995	0.00336737
0.7	200	0.823298	0.823529	0.000231415	0.707108	0.708817	0.00170914
0.7	1000	0.822737	0.823529	0.000792673	0.701275	0.702246	0.000970239
0.7	5000	0.823673	0.823529	0.000143324	0.701141	0.700546	0.000595172
0.9	50	0.94789	0.947368	0.000521818	0.909245	0.908998	0.000246889
0.9	200	0.948035	0.947368	0.000667024	0.903768	0.902939	0.000828809
0.9	1000	0.94798	0.947368	0.000612057	0.901789	0.900749	0.00104031
0.9	5000	0.947294	0.947368	7.41732e-05	0.900004	0.900182	0.000178222

Aggregate MAR validation:

- Median absolute F1 bias: `7.298e-04` (max `8.603e-03`)
- Median absolute AUPR bias: `1.346e-03` (max `2.542e-02`)

This supports the formulas as accurate expectation-level approximations under MAR.

5.6 Non-MAR adversarial missingness

Table 6. Mean deviation from MAR theory by missingness mode

mode	f1_delta_mean	aupr_delta_mean
mar	-0.000259052	-0.00231265
optimistic	-0.000524601	0.0897962
pessimistic	-0.000879459	-0.0733929

AUPR is strongly sensitive to non-MAR labeling structure, while F1 shifts are smaller in this setup.

5.7 External sanity checks

Table 7. Sweep global AUPR values

artifact	candidate_edges	aupr
omnipath	162835	0.000517499
omnipath_dorothea_union_immune_hpn	870806	0.00127928
omnipath_dorothea_intersection_immune_hpn	29253	0.0251236
regulatory	43198	0.00859106

Table 8. Top cross-eval AUPR entries

artifact	system	calibration	candidate_edges	positives	observed_baserate	aupr
cross_intersection	hpn	default	29253	23	0.000786244	0.00373156
cross_intersection_scaled	hpn	default	29253	23	0.000786244	0.00350346
cross_union_scaled	hpn	default	870806	123	0.000141248	0.000677197
cross_union	hpn	raw	870806	nan	nan	0.000674443
cross_union	hpn	logistic	870806	nan	nan	0.000674443
cross_union	hpn	isotonic	870806	nan	nan	0.000645292

These external artifacts are directionally consistent with the ceiling-aware framing: absolute AUPR varies strongly by candidate regime and coverage context.

6. Figures

Figure 1. Theoretical ceilings

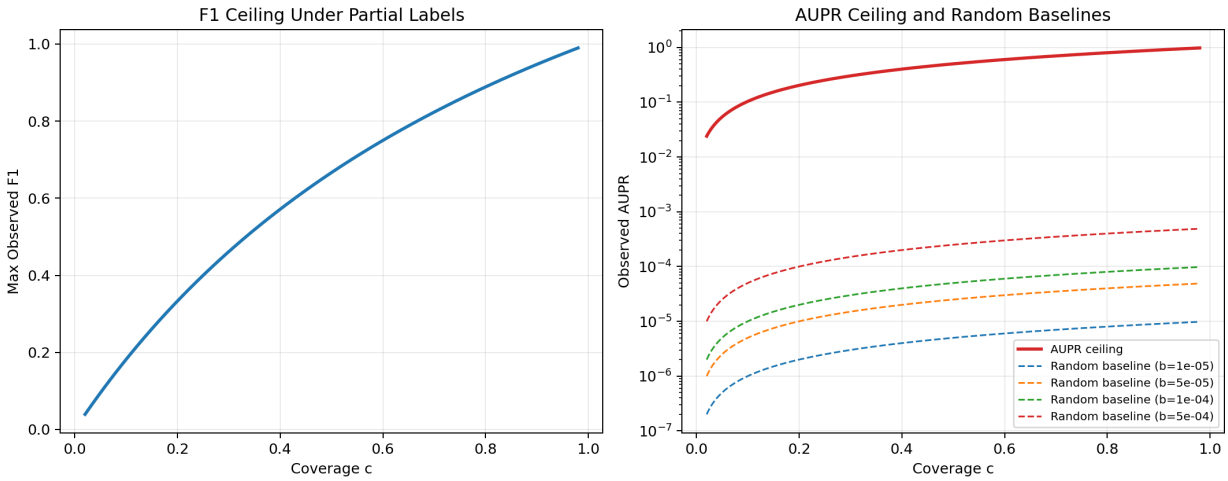


Figure 2. Observed metrics vs ceiling ratios

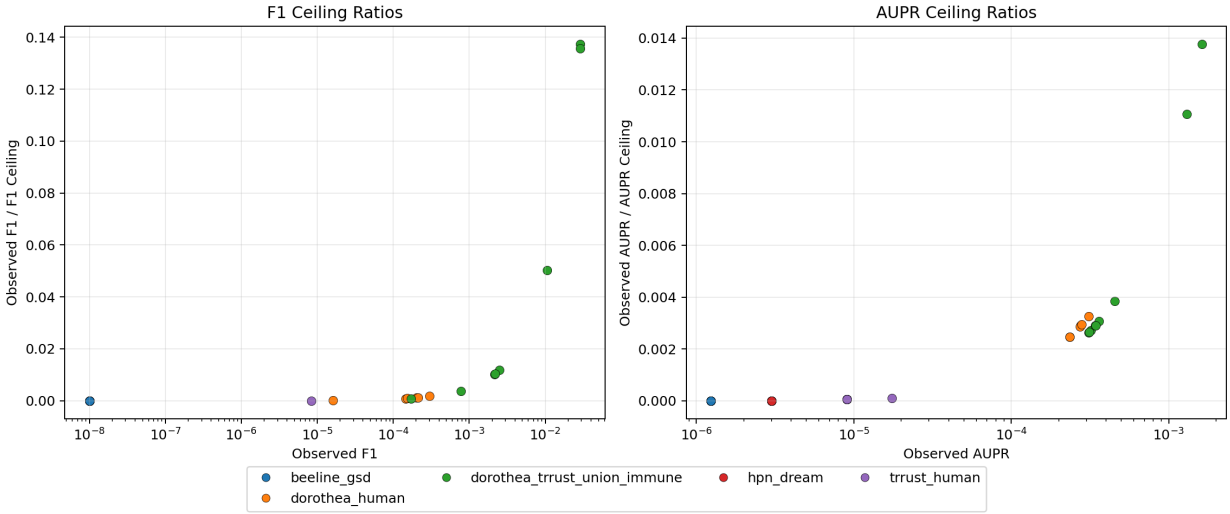


Figure 3. Rank shifts after normalization

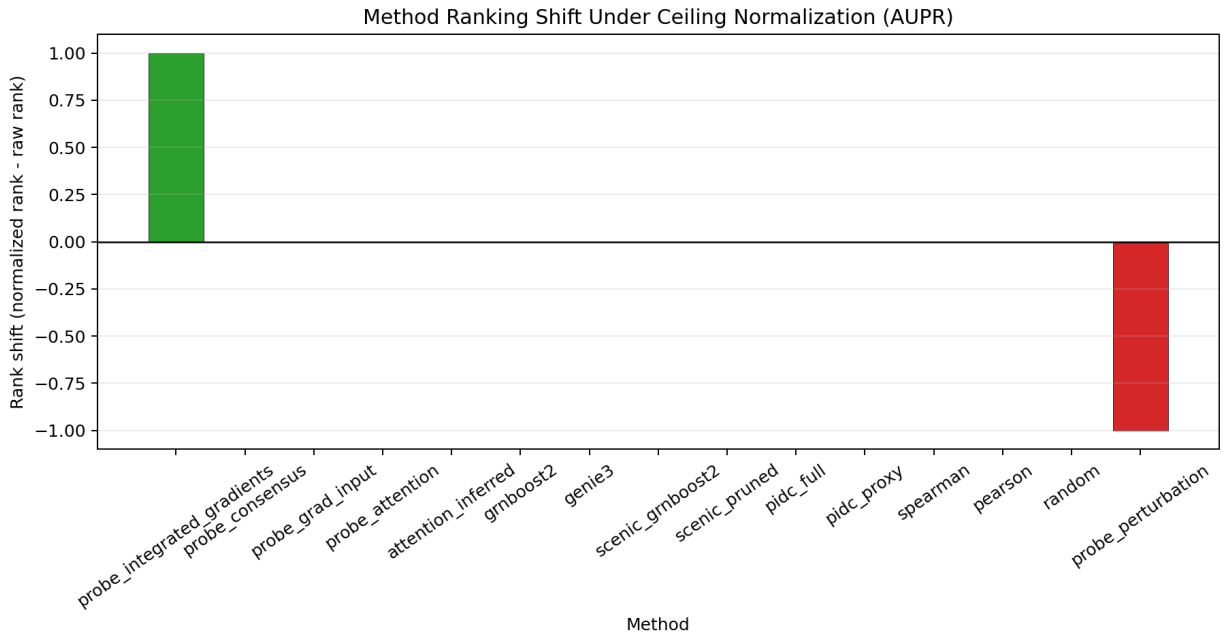


Figure 4. Scenario-specific method ranks

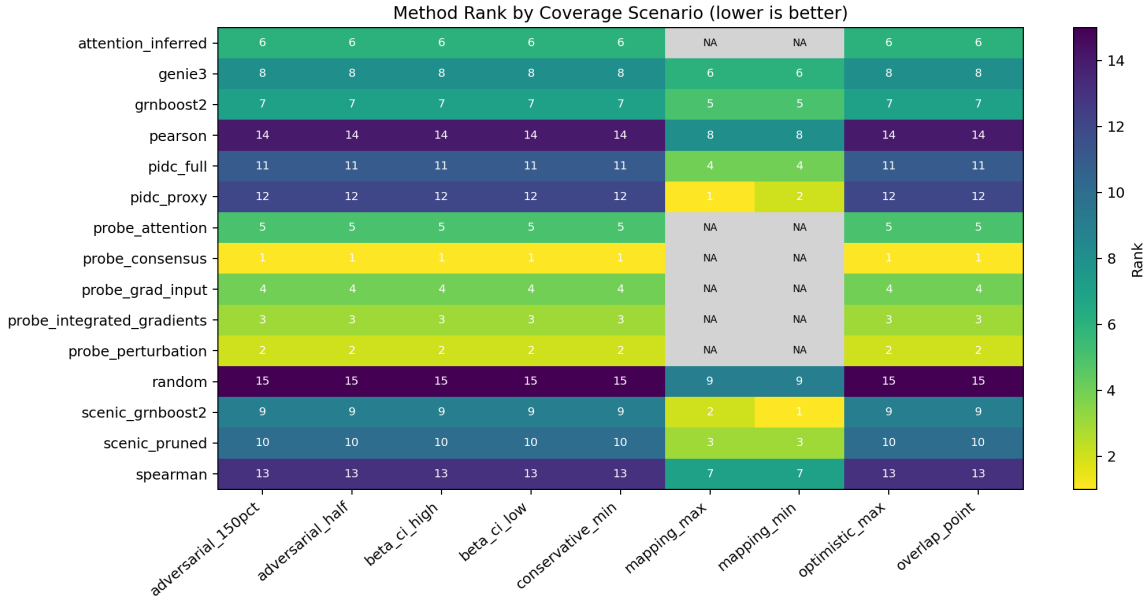


Figure 5. Coverage sensitivity bands

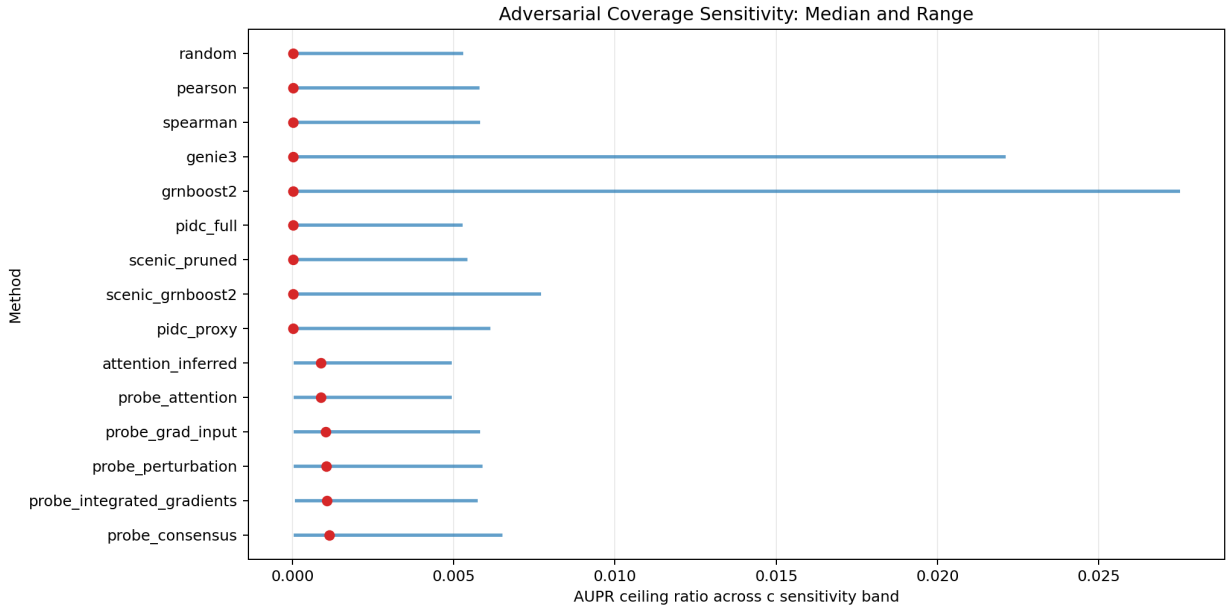


Figure 6. MAR-theory validation

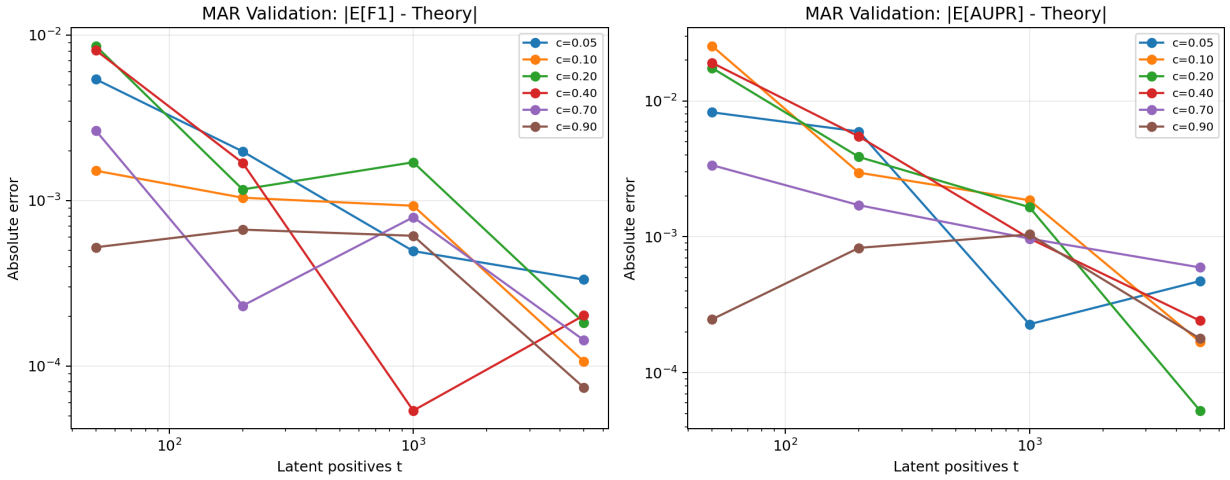
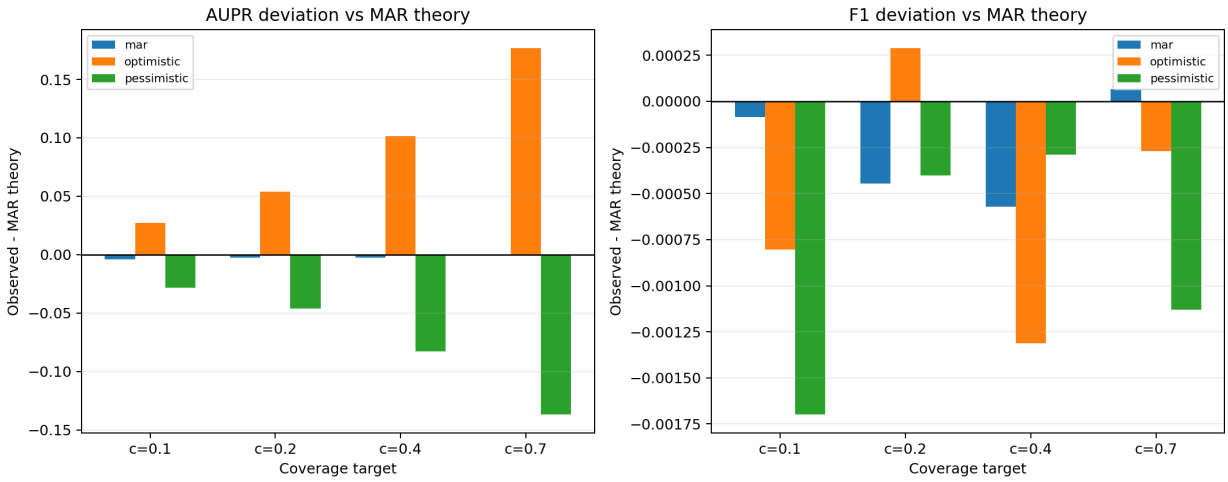


Figure 7. Non-MAR adversarial effects



7. Discussion

Main takeaway: low absolute metrics are not self-explanatory. Ceiling-aware interpretation separates two effects:

1. **Reference incompleteness** (what is fundamentally observable),
2. **Method headroom** (distance from the observable ceiling).

In these results, the second term dominates: observed scores are far below estimated ceilings for most methods/reference pairs.

Adversarial perspective:

- Coverage misspecification can alter normalized rankings, especially when proxy families disagree.
- Non-MAR missingness can substantially bias AUPR relative to MAR predictions.

Therefore, reporting should include both assumptions and sensitivity bands, not single-point ceilings.

8. Limitations

- `c` is proxy-estimated, not identified.
- Method/reference coverage is heterogeneous across artifacts.
- Non-MAR simulations are stylized stress tests, not direct biological curation models.

9. Reproducibility

Run full pipeline:

```
python market_research/ambitious_paper_questions/proposal_01_partial_label_metric_ceilings/scripts/run_
python market_research/ambitious_paper_questions/proposal_01_partial_label_metric_ceilings/scripts/buil
```

Core outputs:

- `outputs/paper/results_summary.md`
- `outputs/paper/tables/*.csv`
- `outputs/paper/figures/*.png`
- `paper/proposal1_partial_label_metric_ceilings_FINAL.md`

10. Appendix: Proof Sketches

A.1 Proposition 1

Linearity of expectation on Bernoulli-thinned positives gives `E[|P intersect L+|] = c|P intersect T|`, yielding the precision/recall relations.

A.2 Proposition 2

For ideal latent predictions, observed recall is 1 and observed precision is `c`, so F1 reduces to `2c/(1+c)`.

A.3 Proposition 3

Under ideal ordering, AP is mean precision at observed-positive ranks among top `t` positions. Taking expectation under Bernoulli thinning introduces the harmonic correction `H_t/t`.